

Bank Stress Testing Toolkit

Macro stress testing with panel econometrics, model validation, and capital projection

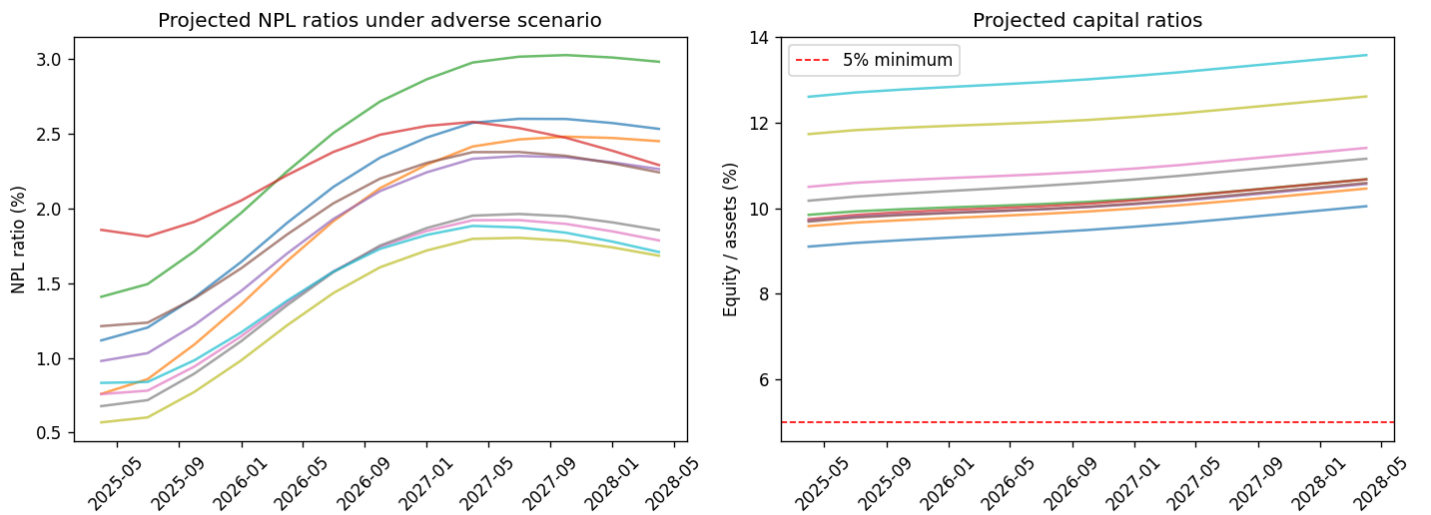
A transparent end-to-end macro stress testing pipeline in Python. The toolkit pulls real US bank financials from the FDIC public API, estimates a panel satellite model linking credit quality to macro variables, validates it with in-sample backtests and out-of-sample forecasts, and projects bank-by-bank NPL paths, credit losses, and capital ratios under Federal Reserve supervisory scenarios.

Built by a former central bank stress tester. At the State Bank of Pakistan I designed and ran quarterly macro stress tests across 52 institutions (USD 220B in assets). This toolkit reimplements the core supervisory architecture on public US data with scenario design, satellite modeling, and dynamic balance-sheet projection as separate inspectable stages.

At a glance

Item	Detail
Sample	10 major US banks, 80 quarters (2005 Q1 to 2024 Q4)
Credit metric	NPL ratio = noncurrent loans / net loans x 100
Satellite model	Dynamic panel OLS with bank fixed effects, HC1 robust SEs
Final regressors	npl_lag1, real_gdp_growth_lag2, unemployment, mortgage_rate_lag1
In-sample backtest	RMSE 0.31 pp, MAE 0.18 pp (one-step ahead)
Out-of-sample forecast	RMSE 0.17 pp (est. through 2022 Q4, forecast 2023-2024)
Stress scenario	2025 Fed severely adverse, 9 quarters from 2024 Q4

Stress test results under the 2025 Fed severely adverse scenario



real_gdp_growth	A191RL1Q225SBEA	Real GDP growth (q/q, SAAR %)
unemployment	UNRATE	Civilian unemployment rate (%)
t3m	TB3MS	3-month Treasury bill (%)
mortgage_rate	MORTGAGE30US	30-year fixed mortgage rate (%)
cpi_inflation	CPIAUCSL (YoY)	CPI inflation (%)
house_price_index	CSUSHPINSA	Case-Shiller national HPI

The panel spans 2005 Q1 to 2024 Q4 (80 quarters x 10 banks = 800 rows, 780 observations after two-quarter lags). The default stress run applies the 2025 Fed severely adverse scenario starting 2025 Q1.

Satellite model

Econometric specification

$NPL(i,t) = a_i + \rho * NPL(i,t-1) + b1 * GDP(t-2) + b2 * UR(t) + b3 * Mtg(t-1) + e(i,t)$, with bank fixed effects a_i . Estimation uses pooled OLS with bank dummies and HC1 heteroskedasticity-robust standard errors. Bank fixed effects only, no time fixed effects, so common macro shocks enter through the regressors.

Term	Variable	Interpretation
a_i	Bank FE	Institution-specific credit culture and portfolio mix
$\rho * NPL(t-1)$	npl_lag1	Credit quality persistence
$b1 * GDP(t-2)$	real_gdp_growth_lag2	Real-economy channel with 2Q lag
$b2 * UR(t)$	unemployment	Labor-market stress (contemporaneous)
$b3 * Mtg(t-1)$	mortgage_rate_lag1	Housing and debt-service channel

Model development

- Baseline: lagged NPL plus GDP, unemployment, and short rate with two lags each.
- GDP transmission: two-quarter GDP lag improved fit and signed consistently with theory.
- Unemployment: contemporaneous term outperformed lagged unemployment in forecast stability.
- Mortgage rate: mortgage_rate_lag1 captured housing credit stress better than T-bill alone.
- Parsimony: dropped insignificant lags and macro variables to limit overfitting on 10 banks.

Estimated coefficients (full panel, 780 obs.)

Variable	Coef.	Std. err.	Sign	Interpretation
npl_lag1	0.942	0.011	+	High persistence
real_gdp_growth_lag2	-0.007	0.002	-	Weaker growth raises NPL with 2Q lag
unemployment	0.052	0.010	+	Higher unemployment raises NPL
mortgage_rate_lag1	0.077	0.008	+	Higher mortgage rates raise NPL with 1Q lag

R-squared = 0.972. Adj. R-squared = 0.972. All coefficients significant at the 1% level.

Model validation API

For model validation roles the toolkit exposes `validate_model()` which returns a `ValidationReport` with diagnostics, backtesting, optional out-of-sample metrics, and PASS/WARN/FAIL flags for challenger-model documentation.

```
from stresskit import SatelliteNPLModel, validate_model

model = SatelliteNPLModel().fit(estimation_panel)
report = validate_model(model, full_panel,
    est_end='2022-12-31', oos_start='2023-03-31', oos_end='2024-12-31')
report.print_summary()
report.save('data/')
report.plot('docs/model_diagnostics.png')
```

Current validation summary: 13 PASS, 5 WARN, 0 FAIL.

Check	Category	Status	Detail
sign_npl_lag1	economic_plausibility	PASS	coef=0.9396, expected +

significance_npl_lag1	statistical	PASS	p-value=0.0000
sign_real_gdp_growth_lag2	economic_plausibility	PASS	coef=-0.0065, expected -
significance_real_gdp_growth_la	statistical	PASS	p-value=0.0002
sign_unemployment	economic_plausibility	PASS	coef=0.0502, expected +
significance_unemployment	statistical	PASS	p-value=0.0000
sign_mortgage_rate_lag1	economic_plausibility	PASS	coef=0.0962, expected +
significance_mortgage_rate_lag1	statistical	PASS	p-value=0.0000
r_squared	statistical	PASS	R ² =0.9715
durbin_watson	serial_correlation	WARN	DW=1.1182
ljung_box	serial_correlation	WARN	residual autocorrelation detected
residual_normality	distributional	WARN	Jarque-Bera p=0.0000
heteroskedasticity	distributional	PASS	Breusch-Pagan p=0.0000, HC1 robust SEs used in estimati
multicollinearity	statistical	WARN	high VIF (>10): unemployment
in_sample_rmse	backtesting	PASS	RMSE=0.3204 pp, MAE=0.1915 pp
gfc_period_rmse	backtesting	WARN	2008-2009 RMSE=0.7491 pp
out_of_sample_rmse	backtesting	PASS	OOS RMSE=0.1726 pp (0.54x in-sample)
out_of_sample_bias	backtesting	PASS	OOS bias=0.1445 pp

Validation results

1. In-sample one-step backtest (full panel)

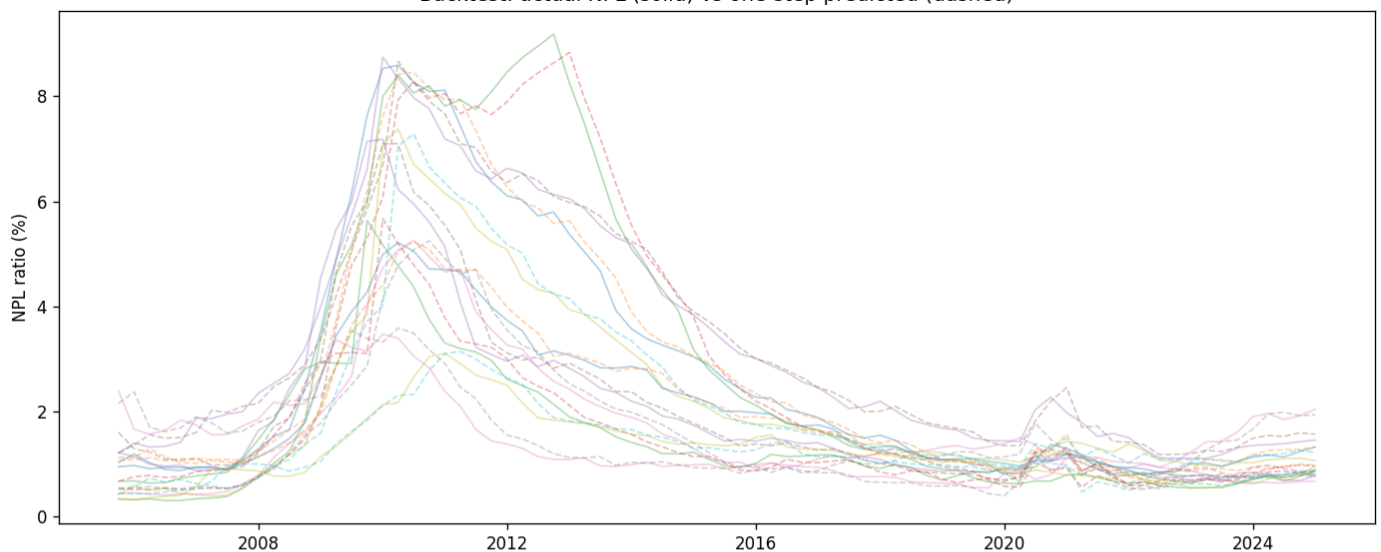
SatelliteNPLModel.backtest() generates one-step-ahead predictions at each quarter t using realized macro data and lagged NPL through $t-1$.

Metric	Value
RMSE	0.31 pp
MAE	0.18 pp

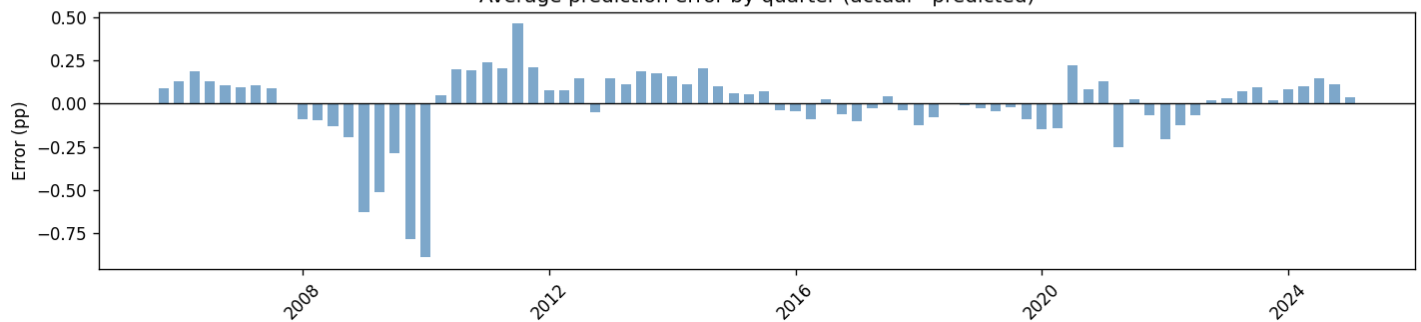
Bank	RMSE	MAE
Regions	0.14	0.11
U.S. Bank	0.19	0.13
Fifth Third	0.23	0.18
Truist	0.24	0.16
Wells Fargo	0.33	0.19
JPMorgan Chase	0.34	0.20
Citibank	0.35	0.21
KeyBank	0.35	0.18
Bank of America	0.40	0.26
PNC	0.40	0.18

Backtest: actual vs one-step predicted NPL

Backtest: actual NPL (solid) vs one-step predicted (dashed)



Average prediction error by quarter (actual - predicted)



2. Out-of-sample forecast (2023 Q1 to 2024 Q4)

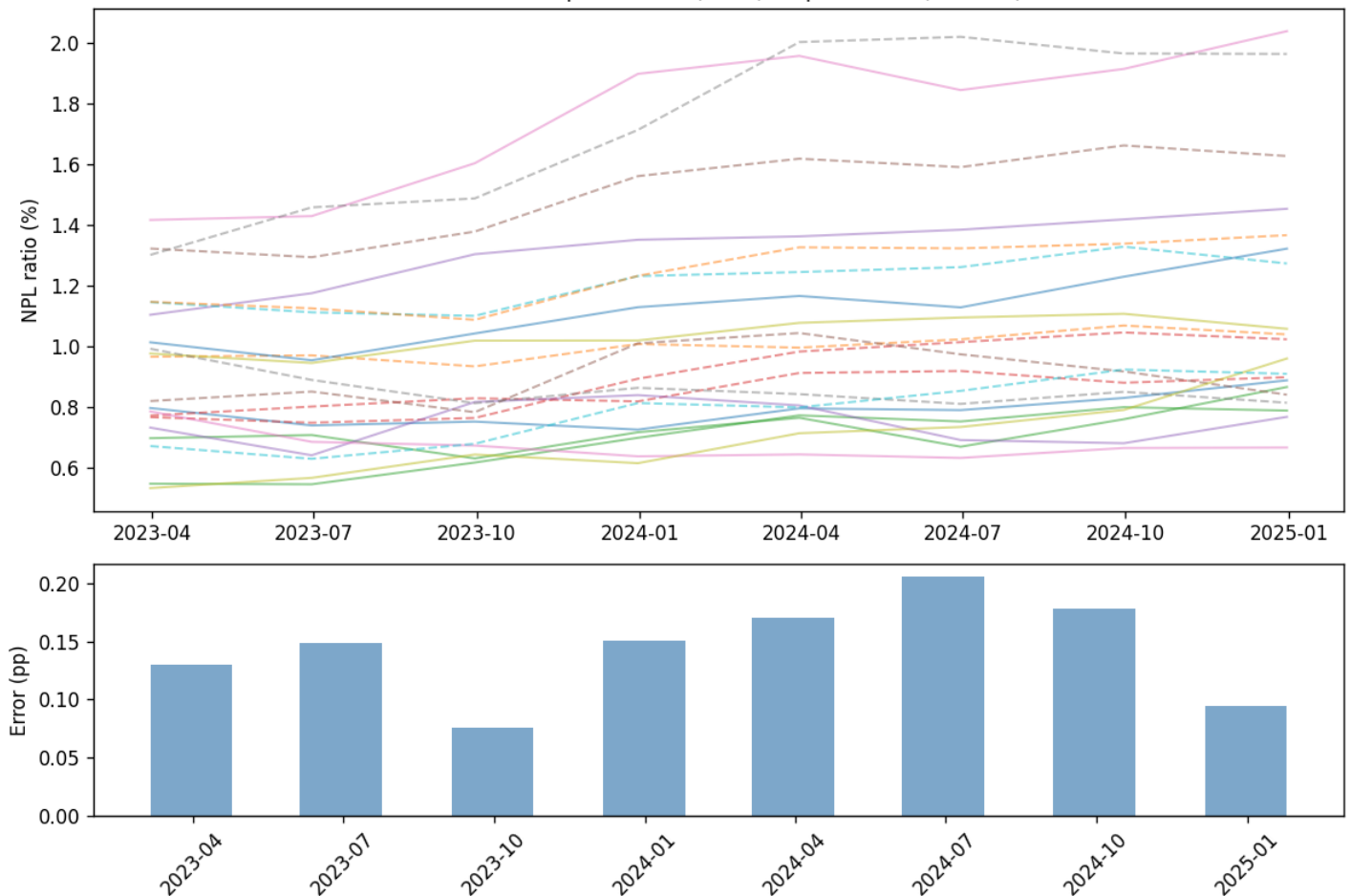
Model estimated through 2022 Q4 only. Forecasts evaluated on 2023 Q1 to 2024 Q4 (80 bank-quarters). Lags are computed on the full panel before slicing the forecast window.

Metric	Value
RMSE	0.17 pp
MAE	0.16 pp
Bias	+0.14 pp (model under-predicts realized NPL)

Quarter	Actual NPL (%)	Predicted (%)	Error (pp)
2023 Q1	0.86	0.99	+0.13
2023 Q2	0.84	0.99	+0.15
2023 Q3	0.91	0.98	+0.08
2023 Q4	0.96	1.11	+0.15
2024 Q1	1.01	1.18	+0.17
2024 Q2	0.97	1.18	+0.21
2024 Q3	1.02	1.20	+0.18
2024 Q4	1.08	1.17	+0.09

Out-of-sample forecast sample

Forecast sample: actual (solid) vs predicted (dashed)



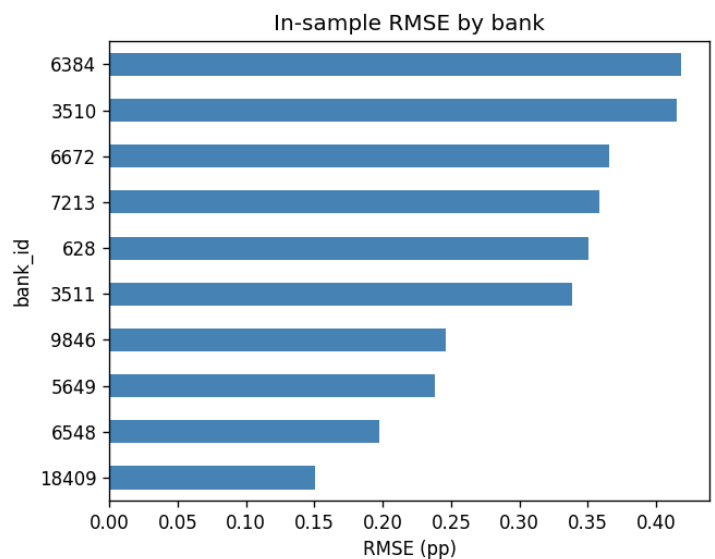
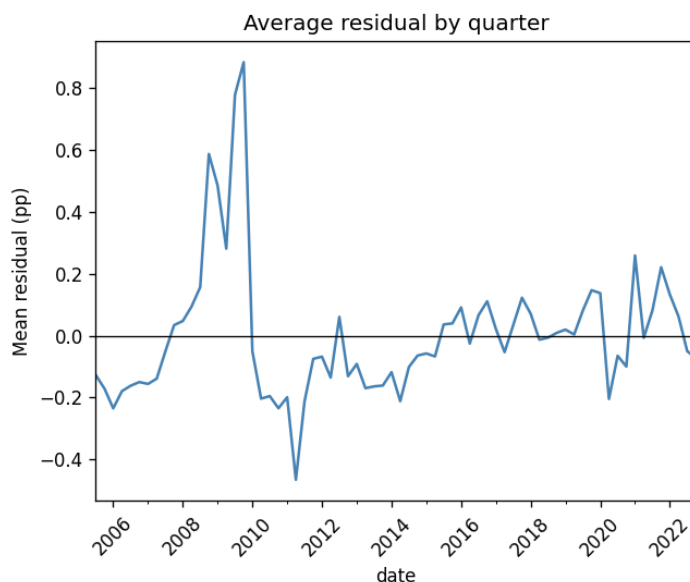
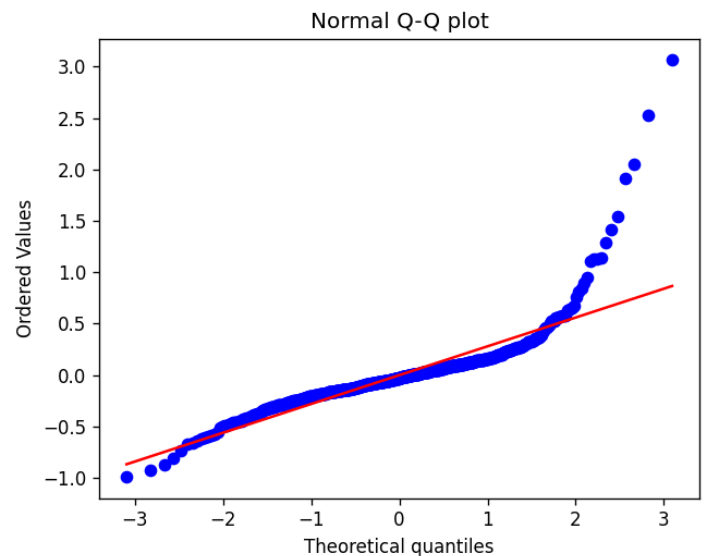
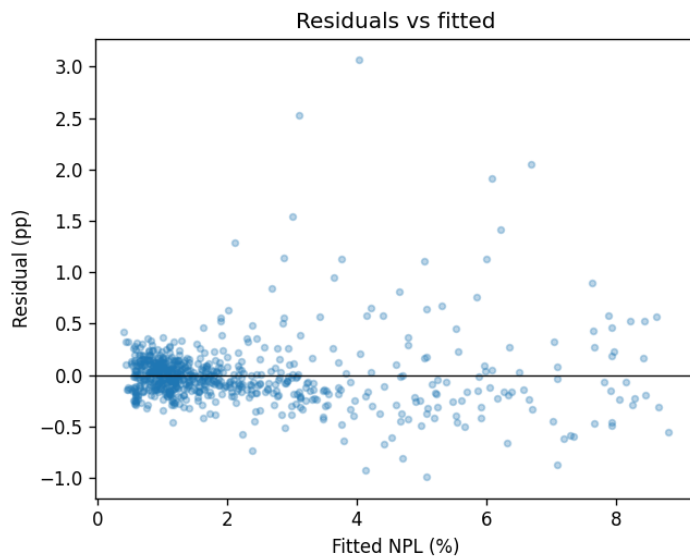
3. Regression diagnostics (estimation through 2022 Q4)

Test	Result	Implication
Durbin-Watson	1.12	Positive residual autocorrelation
Ljung-Box (4/8/12)	$p \sim 0$	Serial correlation in residuals
Jarque-Bera	$p \sim 0$	Non-normal residuals, fat tails from GFC
Breusch-Pagan	$p \sim 0$	Heteroskedasticity, HC1 robust SEs used

Variable	VIF
npl_lag1	3.9
real_gdp_growth_lag2	1.1
unemployment	10.7
mortgage_rate_lag1	6.4

Period	RMSE (pp)	Comment
2005-2007	0.20	Pre-crisis calm
2008-2009	0.75	GFC, model misses crisis nonlinearity
2010-2014	0.29	Post-crisis repair
2015-2019	0.11	Low-volatility credit cycle
2020-2022	0.18	COVID shock partially captured

Model diagnostic charts



4. Unit root tests

Stationarity properties inform satellite model specification. Tests use the arch package.

Series (Phillips-Perron)	Stationary at 5%?
Bank NPL ratios (all 10 banks)	No
real_gdp_growth (and lags)	Yes
unemployment	No (borderline at 10%)
t3m	No

Series (Zivot-Andrews)	Break	Stationary?
BofA NPL	~2007 Q4	No
JPMorgan NPL	~2007 Q4	Yes
real_gdp_growth	~2007 Q4	Yes

NPL ratios are persistent and near unit-root. The AR(1) term in the satellite model is econometrically appropriate. First-differencing would remove level information that stress testers need for adverse scenario projections.

Stress test results

Under the 2025 Fed severely adverse scenario from 2024 Q4 starting positions. NPL ratios roughly double over nine quarters. No bank breaches the 5% equity/assets minimum.

Bank	Trough cap. (%)	End cap. (%)	Credit loss	Breach?
JPMorgan Chase	9.10	10.04	\$10.3B	No
Bank of America	9.58	10.46	\$8.4B	No
U.S. Bank	9.68	10.59	\$2.0B	No
PNC	9.71	10.56	\$1.9B	No
Fifth Third	9.74	10.67	\$0.4B	No
Wells Fargo	9.84	10.67	\$6.3B	No
Citibank	10.17	11.15	\$3.7B	No
KeyBank	10.50	11.41	\$0.6B	No
Truist	11.73	12.61	\$1.7B	No
Regions	12.60	13.57	\$0.8B	No

Parameter	Stress	Default	Notes
lgd	0.45	0.45	Loss given default on new NPLs
ppnr_roa_q	0.10%	0.30%	Quarterly PPNR compressed under stress
min_ratio	5%	5%	Equity/assets breach threshold

Quick start

```
pip install -r requirements.txt
python examples/run_stress_test.py
python examples/run_forecast_sample.py
python examples/run_model_diagnostics.py
python examples/run_phillips_perron.py
python examples/run_zivot_andrews.py
```

First run pulls FDIC data over the internet and caches locally. Subsequent runs use the cache unless the target end date extends beyond cached data.

Known limitations

- Capital ratio is equity/assets, not risk-weighted.
- PPNR is a flat ROA ratio under stress. A second satellite for NII is the natural extension.
- No AFS/HTM mark-to-market under the scenario yield curve.
- Loss timing is immediate. Real frameworks separate default flow and provisioning.
- No management actions such as dividend cuts or capital raises.
- Linear satellite model does not capture GFC-style crisis nonlinearities.
- 10-bank sample is for methodology demonstration, not supervisory conclusions.
- Serial correlation in residuals ($DW = 1.12$). Newey-West would be a standard robustness check.
- Unemployment $VIF = 10.7$. Monitor coefficient stability across estimation windows.

Every assumption is a named parameter because in real stress testing the assumptions are the conversation.

Disclaimer

Educational project on public data. This is not supervisory software, contains no non-public information from any institution, and its outputs are not assessments of any real bank's solvency or capital adequacy.

Author

Umar Shinwari. Former Senior Analyst, Financial Stability Department, Central Bank of Pakistan. MMS Financial Mathematics (Systemic Risk), Yale SOM. Published stress testing and climate risk analysis in the State Bank of Pakistan Financial Stability Review (2022, 2023, 2024).